MIL-B-45588A(WC)
14 January 1972
SUPERSEDING
MIL-B-45588(WC)
19 December 1969

MILITARY SPECIFICATION

BREECH MECHANISM ASSEMBLIES FOR CANNONS, 152MM GUN-LAUNCHER: M81E1 AND XM162E1

1. SCOPE

1.1 This specification covers the breech mechanism assemblies for the 152MM Gun-Launcher Cannons M81E1 and XM162E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance, and Disposition of
MIL-L-46150 - Lubricant, Weapons, Semi-fluid (High Load-Carrying Capacity)

Industry

ANSI B46.1-62 - Surface Texture

DRAWINGS

U. S. Army Weapons Command

D11577392 - Shipping Container for Breech Mechanism Assemblies for Cannons, 152MM Gun-LAUNCHER: M81E1 and XM162E1

PUBLICATIONS

U. S. Army Weapons Command

EPL 7986980 - Engineering Parts List for the Breech Mechanism Assembly for Cannon, 152MM Gun-LAUNCHER: M81E1
EPL 11578135 - Engineering Parts List for the Breech Mechanism Assembly for Cannon, 152MM Gun-LAUNCHER: XM162E1
Master LSQAP 11578068 - Master Index of Supplementary Quality Assurance Provisions for Cannon, 152MM Gun-Launcher: M81E1
Master LSQAP 11578233 - Master Index of Supplementary Quality Assurance Provisions for Cannon, 152MM Gun-Launcher: XM162E1

Source: https://assist.dla.mil -- Downloaded: 20260-22T03:28Z
Check the source to verify that this is the current version before use.

PUBLICATIONS (continued)

U. S. Army Weapons Command
DL 7986980 - Index of Inspection Equipment Lists: Breech Mechanism Assembly for Cannon, 152MM Gun-Launcher: M81E1
DL 11578135 - Index of Inspection Equipment Lists: Breech Mechanism Assembly for Cannon, 152MM Gun-Launcher: XM162E1

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Copies of listed Industry Standards are available to military activities from The U. S. Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120. Non-military activities may obtain copies from: American National Standards Institute, Inc., 1430 Broadway, New York, N. Y. 10018)

3.1 First article. When specified in the contract, the contractor shall submit a first article sample. The definition and inspection of the first article sample shall be as specified in 4.4. Unless otherwise specified (see 6.1.1), the first article sample shall be included in the pilot pack (see 5.1).

3.2 Materials and construction. The breech mechanism assemblies shall conform to this specification, to MIL-C-13931, and the drawings listed in Engineering Parts List 7986980 (M81E1) or 11578135 (XM162E1).

3.3 Surface texture. Waviness is permissible on any surface provided it does not exceed twice the roughness-height-rating (r.h.r.) within the specified cut-off width for the r.h.r. established in ANSI B46.1-62 and does not accumulate for more than one-half the tolerance specified for the surface.

3.4 Protective finish limitation. No protective finish shall be applied by an electrolytic process to the chamber after being autofrettaged.

3.5 Operation. The breech mechanism shall operate in any attitude within 35 degrees elevation, 23 degrees depression, and 15 degrees left or right cant and any ambient temperature between -65° F and +140° F.

3.5.1 Electrical operation.

3.5.1.1 Power. The breech mechanism shall operate with applied drive-motor terminal voltages from 18 to 30 volts DC. At 24 volts ± 1 volt and at an ambient temperature of 75° F ± 5° F, with the exception of a transient surge at the initiation of an opening or closing phase, the current shall not exceed 40 amperes. In any 10 complete cycles the current shall not increase more than 5 amperes.

3.5.1.2 Time. At any voltage from 18 to 30 volts DC the time shall not exceed 3.5 seconds for an opening or closing phase nor 6 seconds for a complete cycle. In addition, at 24 volts ± 1 volt DC and at an ambient temperature of 75° F ± 5° F the time shall not exceed 2.8 seconds for an opening or closing phase nor 5 seconds for a complete cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:28Z
Check the source to verify that this is the current version before use.

4.5.2 Manual operation. The maximum input torque at the hand crank to operate the breech mechanism at 0° elevation and between 15° left to 15° right cant shall not exceed 300 pound-inches at ambient temperatures between -65° F and +140° F.

3.5.3 Breech chamber closure. When the breech is closed the breech mechanism shall prevent any movement of the chamber. When the breech is fully open, the breech mechanism shall prevent the chamber from closing the breech as a result of vibration of vehicle operation.

3.5.4 Seals. The seals for the obturator, the ammunition detent, the firing probe, and the check valve(s) shall prevent the escape of gases and powder residue.

3.5.4.1 Obturator seal. The obturator seal shall be readily removable and the sealing surfaces of the mating parts with specified surface textures shall be protected from damage.

3.6 Lubrication. All movable or sliding surfaces except ball screw threads shall be coated with lubricant as specified in MIL-L-46150 before functional testing.

3.7 Scavenger hardware (XM162E1). The scavenger hardware plunger and chamber valves shall operate with gas bursts of 0.35 ± 0.01 seconds duration at a pressure of 1180 ± 20 psi.

3.8 Proof firing.

3.8.1 Material soundness. After proof firing, all parts shall be free from cracks and fractures and shall comply with any soundness requirements specified on the applicable drawing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions shall be as specified in MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4.2)

b. Quality conformance inspection (see 4.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:28Z
Check the source to verify that this is the current version before use.

4.4 First article.

4.4.1 First article sample. A first article sample shall be submitted by the contractor (in accordance with the contract requirements) for examination by or for the Government and tests by or for the Government (see 6.1.1). The sample shall be representative of the production process to be used during quantity production.

4.4.2 First article inspection. The first article sample shall be subjected to a first article initial production test in accordance with all quality conformance inspections and all examinations and tests described in this specification to assure conformance with the requirements of this specification.

a. Failure of the first article sample to meet the requirements as evidenced by the results of the examinations and tests shall be cause for rejection (see 6.1.1). The contractor shall determine the cause of failure and shall take corrective action in current and subsequent production. Resubmission and retest shall be at the option of the Government (see 6.1.1).

b. Disposition of an acceptable first article sample shall be in accordance with the provisions of the contract (see 6.1.1).

4.5 Quality conformance inspection. Breech mechanism assemblies shall be inspected for conformance to requirements of the provisions of this specification, to MIL-C-13931, SQAPs listed in Master LSQAP 11578068 (M81E1) or Master LSQAP 11578233 (XM162E1), and SQAP-APPENDIX-WVT.

4.5.1.1 Final examination. Final examination of each breech mechanism assembly shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, to MIL-C-13931, and as specified in the contract (see 6.1.2).

4.5.1.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.5.2 Inspection equipment. Data lists containing equipment used in the inspection of parts, subassemblies, and assemblies are listed in DL 7986980 (M81E1) or DL 11578135 (XM162E1).

4.5.2.1 Acquisition, maintenance, and disposition. Unless otherwise specified, responsibility for acquisition, maintenance, and disposition of acceptable inspection and test equipment prescribed in the List of Inspection Equipment DL 7986980 (M81E1) or DL 11578135 (XM162E1), and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.2.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.5.3 Function tests. Each breech mechanism shall be subjected to the following tests. Failure of breech mechanism to meet requirements shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:28Z
Check the source to verify that this is the current version before use.

4.5.3.1 Manual actuation. With the assembly in a level position, manually operate the breech a minimum of 5 cycles to check for malfunction.

4.5.3.1.1 Limit switch. When the breech is being opened, the circuit across pins K and N of the main disconnect switch shall be interrupted when the crank is a minimum of 90 degrees from the mechanical stop. When the breech is being closed, the circuit across pins K and L of the main disconnect switch shall be interrupted when the crank is a minimum of 60 degrees from the mechanical stop.

NOTES: a. Satisfactory manual actuation must be accomplished before any electrical actuation.

b. A cycle is defined as the opening followed by the closing of the breech chamber.

4.5.3.2 Electrical actuation.

4.5.3.2.1 18 volt setting. At an ambient temperature of 75° F ± 5° F with 18 volts + 2, -0 volts DC applied at the launcher harness connection, cycle the breech assembly three times in each position ("A" thru "C"). Record motor current, voltage, and time required for each cycle.

NOTES: a. Electrical cycle actuation rate. During all electrical cycle actuation, a maximum of six complete cycles per minute shall not be exceeded.

b. Delay. After the unit is energized to open the breech, a minimum time delay of 2.0 seconds shall exist prior to energizing the unit to close breech. This time delay shall exist prior to each opening and each closing portion of any cycle.

c. Breech assembly positions (position accuracy + 2 degrees)

Position "A" - level

Position "B" - 35° elevation and 15° right cant

Position "C" - 23° depression and 15° left cant

4.5.3.2.2 30 volt setting. At an ambient temperature of 75° F ± 5° F with 30 volts + 0, -1 volt DC applied at the launcher harness connection, cycle the breech assembly three times in each position ("A" thru "C"). This voltage setting operation shall be considered satisfactory when no evidence of malfunction is detected. Record motor current, voltage, and time required for each cycle.

4.5.3.2.3 24 volt setting. At an ambient temperature of 75° F ± 5°F with 24 volts ± 1 volt DC applied at the launcher harness connection, cycle the breech assembly five times in each position ("A" thru "C"). Record motor current, voltage, and time required for each cycle. Time for each partial cycle shall not exceed 2.8 seconds either to open or to close the breech chamber. The total actual time for each complete cycle shall not exceed 5.0 seconds. (This does not include cycle delay time). The current shall not exceed 40 amperes under any condition of breech attitude with the exception of a transient surge at cycle initiation or closure.

Source: https://assist.dla.mil -- Downloaded: 20260-22T03:28Z
Check the source to verify that this is the current version before use.

4.5.3.3 Ejection test. Ejection shall be checked with an empty missile aft cap. Ejection shall be considered acceptable if the empty cap is fully moved onto the loading tray where it may be brushed off by hand.

4.5.3.4 Firing circuit resistance test. The breech assembly harness resistance for the firing probe circuit shall be checked. The resistance shall not exceed 250 milliohms measured between harness input terminals "G" and "P" with harness connected to probe.

NOTE: The resistance shall be checked while using a suitable dummy connector to provide a closed circuit across the firing probe. The resistance value limit shall include firing probe assembly resistance only, without resistance of dummy connector or its attachment.

4.5.3.5 Firing circuit insulation resistance test. With the harness assembly installed to all breech electric connections, apply 500 volts DC for one to five seconds to harness input "Pins A and G", with all other leads and breech mechanism simultaneously grounded. The firing circuit shall show a minimum resistance of 100 megohms to ground with the breech closed. Accuracy of this equipment shall be within 5 percent.

4.5.3.6 Harness assembly insulation resistance test. With the harness assembly removed from the breech assembly apply 500 volts DC for one to five seconds to all pins of the harness input connector except "Pin P", with all the connector shells, mounting bracket and mounting block grounded. The harness assembly shall show a minimum resistance of 100 megohms on all wires to each other and to ground.

4.5.3.7 Scavenger hardware test (XM162E1). Warning: Observe appropriate safety precautions while testing with compressed gas. (a) Open the breech mechanism so that the breech chamber is fully opened. With gas bursts of specified pressure and duration, operate both plunger valves a minimum of three cycles. The plunger valve should extend into the interior of the cannon on application of gas pressure and completely retract on removal of gas pressure. Verify by audible and visual means, the operation of the plunger valves. (b) Close the breech mechanism so that the breech chamber is fully closed and locked. With gas bursts of specified pressure and duration, operate both chamber valves a minimum of three times. Verify by audible means the operation of the chamber valves.

4.5.4 Environmental test. Acceptance of breech mechanism assemblies is subject to the successful completion of the following test, required to be run by the supplier of the complete cannon. The first production breech mechanism installed in a gun-launcher and one from each succeeding 100 units shall be tested electrically and manually in a level position "A" at temperatures of +140° F and -65° F and shall operate in accordance with the requirements.

4.5.4.1 The breech assembly shall be operated manually 5 cycles before the electrical operation.

4.5.4.2 With 24 volts $\pm$ 1 volt DC applied at the launcher harness connections, cycle the breech assembly five times. Record motor current, voltage, and time required for each cycle.

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:28Z
Check the source to verify that this is the current version before use.

4.6 Proving ground inspection. Each breech mechanism when assembled to a gun-launcher shall be subject to proving ground inspection at a test agency designated by USATECOM. The inspection shall be performed by Government personnel.

4.6.1 Proof firing. Each breech mechanism shall be subject to proof firing by firing one (1) specific pressure round, which shall produce 113 ± 4 percent of upper pressure limit.

4.6.2 Inspection after proof firing.

4.6.2.1 Material soundness. After proof firing, the breech mechanism shall be partially disassembled to permit removal and magnetic-particle or liquid penetrant testing of the breech chamber and coupling in accordance with the applicable drawing. The remaining assemblies shall be free from cracks or fractures as evidenced by visual examination.

4.6.2.2 Seals. Examine obturator seal, ammunition detent seal, firing probe seal, check valve seal, and adjacent areas of mating parts for evidence of gas leakage during proof firing.

4.6.2.3 Breech chamber closure. The breech mechanism, when assembled in the vehicle, shall be operated under specified conditions. The breech mechanism shall be inspected to determine compliance with 3.5.3.

4.6.2.4 Proof marking. Proof acceptance marking shall be located on the breech mechanism near the serial number.

4.7 Inspection of preparation for delivery.

4.7.1 Classification of defects. The classification of defects and AQLs for determining compliance with the requirements for preparation for delivery are as follows:

Category	Defect	AQL	Inspection method
Major			
101	Illegible or incorrect marking	1.0	Visual
102	Improper packaging or packing	1.0	Visual
103	Inadequate cleaning or drying	1.5	Visual
104	Improper application of preservative	1.5	Visual
105	Improper or inadequate blocking or bracing	1.5	Visual/ Manual
106	Improper closure or strapping of containers	1.5	Visual
Minor			
201	Workmanship	4.0	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:28Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract, a pilot pack shall be forwarded in accordance with 3.1. Pilot packs for the breech mechanism shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of Watervliet Arsenal Drawing D11577392.

5.2 Preservation, packaging, and packing. The breech mechanism shall be unit packaged, packed, and marked in accordance with the requirements of Watervliet Arsenal Drawing D11377392.

5.3 Repair parts. Repair parts shall be unit packaged, packed, and marked in accordance with the requirements of the applicable packaging data sheet (see 6.1.1).

6 NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and pilot pack, if different (see 3.1 and 5.1)

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should designate at whose expense a retest may be performed.

(5) The contract should specify that the acceptable first article sample shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification shall be specified and identified as a separate item of the contract.

c. Serial numbers for the breech mechanisms

d. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.2.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:28Z
Check the source to verify that this is the current version before use.

e. Extent of supplier's responsibility for Government-furnished equipment

f. Selection of applicable levels of preservation, packaging, and packing (see 5.2)

g. Packaging instructions for repair parts (see 5.3)

h. Applicable acceptance test procedures and frequency of testing (see 4.6)

6.1.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be specified for delivery on DD Form 1423 in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.3 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

6.4 Frequency of testing is defined as the proportion of the lot to be tested after initial 100 percent testing.

Custodian:
Army--WC

Preparing activity:
Army--WC

Project No. 1025-A016

Source: https://assist.dla.mil -- Downloaded: 202640-22T03:28Z
Check the source to verify that this is the currentversion before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

To detach this form, cut along this line.

Source: https://assist.dla.mil -- Downloaded: 202610-22T03:28Z
Check the source to verify that this is the current version before use.